

SAFETY DATA SHEET**1.CHEMICAL PRODUCT AND COMPANY IDENTIFICATION**

Product Name	PENGUIN CEMENT SH-301H
Manufacturer Name	Sunstar Chemical (Thailand) Co.,Ltd.
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2.HAZARDS IDENTIFICATION**PHYSICAL HAZARDS**

Not classified

HEALTH HAZARDS

Acute toxicity (Oral)	Not classified
Acute toxicity (Dermal)	Category 4
Acute toxicity (Inhalation: Gases)	Not applicable
Acute toxicity: (Inhalation: Vapours)	Classification not possible
Acute toxicity (Inhalation: Dusts and mists)	Classification not possible
Skin corrosion/irritation	Category 2
Serious eye damage/eye irritation	Category 2A
Respiratory sensitization	Classification not possible
Skin sensitization	Category 1
Germ cell mutagenicity	Category 2
Carcinogenicity	Category 1
Reproductive toxicity	Not classified
Specific target organ toxicity - Single exposure	Not classified
Specific target organ toxicity - Repeated exposure	Category 1 (Respiratory system)
Aspiration hazard	Classification not possible

ENVIRONMENTAL HAZARDS

Hazardous to the aquatic environment (Acute)	Category 1
Hazardous to the aquatic environment (Long-term)	Category 1
Hazardous to the ozone layer	Classification not possible

ALLOCATION OF LABEL ELEMENTS**Hazard symbol****Signal word**

Danger

Hazard statement

H312 Harmful in contact with skin.
H315 Causes skin irritation.
H317 May cause an allergic skin reaction.
H319 Causes serious eye irritation.
H341 Suspected of causing genetic defects.
H350 May cause cancer.
H372 Causes damage to organs through prolonged or repeated exposure.
H400 Very toxic to aquatic life.
H410 Very toxic to aquatic life with long lasting effects.

Precautionary statements**Prevention**

P201 Obtain special instructions before use.
P202 Do not handle until all safety precautions have been read and understood.
P260 Do not breathe dust/fume/gas/mist/vapours/spray.
P261 Avoid breathing dust/fume/gas/mist/vapours/spray.
P264 Wash thoroughly after handling.
P270 Do not eat, drink or smoke when using this product.
P272 Contaminated work clothing should not be allowed out of the workplace.
P273 Avoid release to the environment.

P280 Wear protective gloves/protective clothing/eye protection/face protection.

Response

P312 Call a POISON CENTER or doctor/physician if you feel unwell.

P314 Get medical advice/attention if you feel unwell.

P321 Specific treatment (see label).

P391 Collect spillage.

P302+P352 IF ON SKIN: Wash with plenty of soap and water.

P308+P313 IF exposed or concerned: Get medical attention/advice.

P332+P313 If skin irritation occurs: Get medical advice/attention.

P333+P313 If skin irritation or rash occurs: Get medical advice/attention.

P337+P313 If eye irritation persists: Get medical advice/attention.

P362+P364 Take off contaminated clothing and wash it before reuse.

P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

Storage

P405 Store locked up.

Disposal

P501 Dispose of contents/container to..... (in accordance with local/regional/national/international regulation).

3.COMPOSITION/INFORMATION ON INGREDIENTS

Substance/Preparation		Preparation
General Name		Epoxy type Adhesives
Components Contributing to the Hazard		
Components	Cas No	%WT
Epoxy resin	25068-38-6	30~40
Calcium oxide	1305-78-8	<5
Carbon black	1333-86-4	<10

4.FIRST AND MEASURES

Inhalation

Remove to fresh air and keep at rest in a position comfortable for breathing.

Skin Contact

Wash with plenty of soap and water.

If skin irritation or rash occurs: Get medical advice/attention.

Eye Contact

Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

If eye irritation persists: Get medical advice/attention.

Ingestion

Get medical attention/advice.

5.FIRE-FIGHTING

Extinguishing Media

Carbon dioxide, dry chemical, water, foam, dry sand

Special Hazards

Toxic gases (nitrogen oxides) will foam upon combustion.

Special Methods

Evacuate non essential personnel.

Move container from fire areas if it can be done without risk.

Keep personnel removed from and upwind of fire.

Protection of Firefighters

Wear self-contained breathing apparatus.

6.ACCIDENTAL RELEASE MEASURES

Personal Precautions

Wear protective equipment and avoid contact with skin and eye.

Environmental Precautions

Avoid release to the environment.

Methods for Cleaning up

Absorb with liquid-binding materials (sand, sawdust etc.).

Scoop material into containers for disposal.

7. HANDLING AND STORAGE

Handling

Technical Measures

Do not handle until all safety precautions have been read and understood.
 Do not breathe dust/fume/gas/mist/vapours/spray.
 Wash thoroughly after handling.
 Do not eat, drink or smoke when using this product.
 Contaminated work clothing should not be allowed out of the workplace.
 Keep containers closed when not in use.

Precautions

Use in the well-ventilated areas.

Storage

Suitable Storage Conditions

Keep container tightly closed.
 Store in a cool (<10°C) and well-ventilated place.
 Protect material from direct sunlight.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

Components	Cas No	ACGIH (2018,TWA)
Epoxy resin	25068-38-6	Not established
Calcium oxide	1305-78-8	2 mg/m ³
Carbon black	1333-86-4	3 mg/m ³

Personal Protective Equipment

Hand Protection

Protective gloves

Eye Protection

Safety glasses

Skin and Body Protection

Avoid contact with skin.

Hygiene Measures

Wash hands before taking a break and at the end of work.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical State

Form	Paste
Color	Black
Odor	Slightly odor
pH	Not established
Flash point	>200°C
Density	About 1.4
Solubility	Not established
Other data	Not established

10. STABILITY AND REACTIVITY

Stability

Stable under the condition of normal usage.

Possible Hazardous Reactions

Reactive with amine compounds

Conditions to Avoid

Avoid amine compounds (amine, amide etc.)

Hazardous Decomposition Products

Nitrogen oxides

11. TOXICOLOGICAL INFORMATION

Acute toxicity

Epoxy resin :	LD50 (Oral) :	Rat	11,400 mg/kg	(CERI Hazard Data 2001-36 (2002))
Calcium oxide :	LD50 (Oral) :	Rat	5,000 mg/kg and 5,916 mg/kg	(Risk Assessment Report (Food Additives) (Food Safety Commission of Japan, 2013))
Carbon black :	LD50 (Oral) :	Rat	>8,000 mg/kg and >10,000 mg/kg	(SIDS (2007))

Skin corrosion/irritation

Epoxy resin :	Category 2
Calcium oxide :	Category 2
Carbon black :	Not classified

Serious eye damage/eye irritation

Epoxy resin :	Category 2B
Calcium oxide :	Category 1
Carbon black :	Not classified

Skin sensitization

Epoxy resin :	Category 1
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Germ cell mutagenicity

Epoxy resin :	Not classified
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Carcinogenicity

Carbon black :	Category 2
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Reproductive toxicity

Epoxy resin :	Not classified
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Specific target organ toxicity - Single exposure

Calcium oxide :	Category 1 (respiratory organs)
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Specific target organ toxicity - Repeated exposure

Calcium oxide :	Category 1 (respiratory organs)
Carbon black :	Category 1 (respiratory system)

12.ECOLOGICAL INFORMATION**Hazardous to the aquatic environment (Acute)**

Epoxy resin :	EC50 (48H) :	Crustacea (Daphnia magna)	1.7 mg/L	(CERI/NITE Hazard Assessment Report (preliminary version) (2006) and others)
Carbon black :	EC50 (72H) :	Algae (Scenedesmus)	>10,000 mg/L	(SIDS, 2007)
	EC50 (24H) :	Crustacea (Daphnia magna)	>5,600 mg/L	(SIDS, 2007)
	LC50 (96H) :	Fish (Dace)	>1,000 mg/L	(SIDS, 2007)

13.DISPOSAL CONSIDERATIONS**Waste from Residues**

Follow the all regulations in your country.

Contaminated Packaging

Follow the all regulations in your country.

14.TRANSPORT INFORMATION

UN Class 9

UN No 3077

Packing group III

Specific Precautionary Transport Measures and Conditions

Follow the all regulations in your country.

15.REGULATORY INFORMATION

Follow the all regulations in your country.

16.OTHER INFORMATION

The information herein is given in good faith, but no warranty, express or implied, is made.

The information presented herein is believed to be accurate but is not warranted. Recipients are advised to confirm in advance that the information is current, applicable and suitable to there circumstances.